50 Java 8 Programs for Interview Preparation

CloudTech • Student Practice Checklist • Problem statements only — no solutions

How to use this checklist

Solve each problem using Java 8 features where appropriate, including Streams, lambdas, method references and Collectors. Consider edge cases, duplicates, nulls, ordering and empty inputs. Difficulty levels are approximate and depend on the chosen approach.

Page 1 — Programs 1–17: Numbers and Strings

No.	Java 8 interview program	Difficulty
1	Print all even numbers from a list	Easy
2	Print all odd numbers from a list	Easy
3	Find the sum of all numbers in a list	Easy
4	Find the average of all numbers in a list	Easy
5	Find the maximum number in a list	Easy
6	Find the minimum number in a list	Easy
7	Find the second-highest distinct number in a list	Medium
8	Find duplicate elements in a list	Medium
9	Find distinct elements from a list	Easy
10	Count the frequency of each element in a list	Medium
11	Sort integers in ascending order	Easy
12	Sort integers in descending order	Easy
13	Remove duplicate elements from a list	Easy
14	Join strings with a delimiter	Easy
15	Find the first non-repeated character in a string	Medium
16	Check whether a string is a palindrome	Easy
17	Convert strings to uppercase	Easy

Practice tip: explain your approach and time/space complexity.

Page 2 — Programs 18–34: Strings, Lists and Collectors

No.	Java 8 interview program	Difficulty
18	Convert strings to lowercase	Easy
19	Convert strings to title case	Easy
20	Find the longest string in a list	Medium
21	Find the shortest string in a list	Medium
22	Sort strings by length	Medium
23	Group strings by length	Medium
24	Find the first string starting with a prefix	Easy
25	Count strings starting with a prefix	Easy
26	Find the top N most frequent elements	Medium
27	Partition numbers into even and odd	Easy
28	Merge two lists and remove duplicates	Easy
29	Find common elements between two lists	Easy
30	Find non-common elements between two lists	Medium
31	Find the intersection of two lists	Easy
32	Find the union of two lists	Easy
33	Convert a list of objects to a map	Medium
34	Group employees by department	Medium

Practice tip: explain your approach and time/space complexity.

Page 3 — Programs 35–50: Employee and Advanced Problems

No.	Java 8 interview program	Difficulty
35	Find the highest-paid employee in each department	Medium
36	Find the second-highest salary among employees	Medium
37	Calculate average salary in each department	Medium
38	Find employees earning above a given amount	Easy
39	Find the oldest and youngest employee	Medium
40	Sort employees by multiple fields	Medium
41	Count employees in each department	Easy
42	Map employees using employee ID as key	Easy
43	Find duplicate employees by a specific field	Medium
44	Remove null values from a list	Easy
45	Find the kth-largest distinct element	Medium
46	Flatten a list of lists	Easy
47	Find the top N highest elements	Medium
48	Check whether two lists are equal	Easy
49	Find prime numbers from a list	Medium
50	Partition numbers into positive and negative values	Easy

Practice tip: explain your approach and time/space complexity.